From image to inspectable candidate fits

Local browser interface → Python analysis → evidence, candidate labels and reproducible exports.

1 Select an image and analysis settings

Browser controls the image/ROI, enabled phases, detector settings, reference cells and display. Requests carry an explicit settings snapshot.

2 Local HTTP service

server.py validates requests, serializes engine access and manages grid jobs, saved entries and exports.

3 Validate and load calibrated pixels

Engine loads finite 2D DM4 or bundled demo arrays. Calibration units choose real-space FFT or direct diffraction.

4 Extract observed profiles → panel 2

FFT: crop, window, transform and detect peaks.

Diffraction: detect spots in recorded pixels; optionally refine one shared beam centre.

5 Compare periodic structures → panels 3–5

Generate allowed HCP/BCC/FCC predictions; fit rotation/scale, assign peaks, score and screen.

A separate freely fitted reciprocal-grid check reports one or two periodic patterns.

6 Return numerical results and display fields

Profiles, candidate models, rank components, geometry, flags, settings, timings and engine hash.

Browser renders overlays, tables, previews and exports.

Image collection

data/ and output/ point to chosen locations; .tem-paths.json records the choices.

DM4 metadata supplies calibration. Bundled NPZ/JSON demos include pixel checksums.

Material and structure inputs

materials.py: editable nominal cells and phase defaults.

templates.json + crystal_templates.py: a finite set of zone axes and reflection rules.

Interpretation

Candidate labels are not confirmed phases.

Display is separate from fitting

Palette, contrast, marker shape/motion and Raw FFT / Processed view do not rerun fitting.

Processed view exposes the smoothed background-subtracted signal; the original FFT remains available.

Many ROIs → panel 6

A background grid job calls the same analysis for each cell. Adaptive refinement decides which children replace a parent.

Current implementation: sequential cell analysis. SciPy FFT uses two workers; grid parallelization is not implemented.

Keep and reproduce

Browser storage: basic preferences.

Project: named ROIs, source identities, tags and settings. Exports: numerical JSON and separate PNG/GIF visuals.

Two routes to measured reciprocal-space profiles

Coordinates are in nm^{-1} (cycles per nm, without a 2π factor). Detection uses calibrated source pixels.

REAL SPACE / ROI → FFT

z = (I − mean(I)) × W; mean removal and Hann W are optional.

L = log(1 + |fftshift(FFT$_2$(z))|^2). Frequency bins follow the crop: Δg = max[1/(w·sx), 1/(h·sy)]. Display spans ±1.05·gmax.

Remove radial background and smooth

Bin radius by floor(|q|/Δg); subtract the mean L in each radial bin.

D = Gaussianσ(L − radial_mean). Default σ = 0.65 FFT bins. The Raw FFT / Processed toggle shows L or D with the same coordinates.

Threshold connected profiles

Within gmin ≤ |q| ≤ gmax: T = max(Tfloor, f·max(D)).

Default Tfloor = 1.8; f = 0.35. One maximum per connected component of D > T; require maximum ≥ minimum strength.

Auto strength searches subsets of this pool (panel 4). It does not lower T or change σ.

Refine positions and keep independent evidence

Parabolic x/y refinement, limited to ±0.5 bin. Fold q and −q to one profile; strongest-first merge below the larger of the user separation and window width.

Default cap: 70 profiles. Weight w = clip(Dpeak − 2, 1, 6).

FFT matching tolerance

Approximate window power width: 1.44·Δg (Hann), 0.886·Δg (rectangular crop).

Matching radius τ = max(0.10 nm^{-1}, window width). Default fitting band is 3–10 nm^{-1}, adjustable. Full outer-band Nyquist coverage is required for phase fitting.

DIRECT DIFFRACTION / detector → coordinates

No FFT: qx = (x − cx)·sx; qy = (cy − y)·sy.

Initial centre: clusters of opposite-spot midpoints, with a central prior and beam-stop shadow cue; detector midpoint is the labelled fallback.

Find spots across position and size

Masked Gaussian differences at σ and 2σ; check successive scales (2, 4, 8, 16, 32 detector pixels by default).

Find spatial/scale maxima; compare filtered response AND native intensity against local annular background.

Local contrast, centroid and deduplication

Background annulus: 4σ–8σ. Strong spots must exceed the user contrast cutoff in both response and intensity (default 9).

Positive-core centroids and spot widths refine positions. Strongest-first suppression merges nearby detections of the same spot across scales.

Optional shared translation refinement

At least 3 opposite pairs spanning directions: robust least squares fits their common midpoint, within the shift limit (default 50 px).

Redetect once if the centre changes. Reject inconsistent/competing origins; HCP, BCC and FCC share the same centre.

Direct-spot profiles and tolerance

Group observed opposite spots once; never invent a missing partner. Weight w = clip(log(1 + contrast), 1, 6); default cap 70.

τ = max(0.10 nm^{-1}, estimated spot width). Default band: 2.5–10 nm^{-1}; centre/masks exclude obscured areas. Only observed spots are marked.

Generate, rotate and scale allowed patterns

Predictions move to match observations. Image pixels, reciprocal axes and measured peak coordinates do not stretch.

1 Reference unit cells

Material presets supply editable a_0 and c_0. Cubic: a = b = c. HCP: a = b, γ = 120°.

Only enabled phases enter the search. Zone directions form a finite library; this is not an exhaustive orientation search.

2 Enumerate integer reflections

Cubic: g(hkl) = (h, k, l) / a.

HCP: g(hkl) = ((2h+k)/(√3·a), k/a, l/c). Internal indices use three coordinates.

BCC: h+k+l even. FCC: h,k,l all even or all odd. HCP: exclude odd l when (h−k) is divisible by 3.

Keep zone-law hu+kv+lw = 0; project onto its reciprocal plane.

3 Supply enough reciprocal nodes

Integer bounds depend on reference dimensions, outer radius and smallest allowed scale; no fixed shell count.

Each scored prediction is filtered again to the current radial band. q and −q represent one profile.

Allowed does not mean visible

These rules generate positions, not simulated intensities. Thickness, excitation, contrast transfer and masking can hide reflections.

No multislice, dynamical-intensity, ring-indexing or stacking-fault model is used.

4 Apply a physically defined metric

Cubic / uniform mode: one reciprocal multiplier s. HCP dual mode: sa and sc where c is visible. Bounds: 0.85–1.15.

For plane basis P and D = diag(sa,sa,sc), use the plane metric $M = P^T D^2 P$ and transform $\sqrt{M}$.

Model dimensions: $a = a_0/sa$; $c = c_0/sc$.

5 Seed orientations and scales

Use up to 16 strongest observed profiles to seed in-plane rotation and reciprocal scales.

HCP dual search scans 11 c*/a* ratios plus 1, then continuous refinement. Fast nearest-neighbour quality must reach 0.20.

Retain up to 4 dual-scale or 3 single-scale seeds per template.

6 Alternate assignment and refinement

Assign peaks one-to-one (right), then solve weighted positional least squares for rotation and visible scales.

Up to 3 rounds; rotation moves at most ±0.06 rad per round. Accept a new solution only if its position score does not decrease.

Scale penalties are applied after this positional optimization.

7 Retain distinct fits

Rank by the adjusted score (panel 4). Keep up to 5 per phase; suppress near-identical profile assignments within the same zone in automatic mode.

One-to-one profile assignment

For observed oi and predicted pj:

dij = min(||oi − pj||, ||oi + pj||)

Qij = exp[−½(dij/(τ/2))²] if dij ≤ τ; otherwise Qij = 0.

The same tolerance is used for every structure.

Hungarian assignment

Minimize costs −wi·Qij, with zero-cost dummy columns for unmatched observations.

A measured profile can support only one node within a candidate; a prediction cannot claim several profiles.

Matched opposite spots count once.

Fixed-scale path

Stored/custom scale uses fixed dimensions. Seed rotations at 0.5° increments; refine rotation twice, capped at 2° per step.

The best 2 quick seeds per template are evaluated. Fixed-mode retention suppresses duplicate support within a phase, even across equivalent zones.

No automatic scale cost applies.

Search limits

Finite zone library and seed budget: “best” means best retained candidate found by this search, not a guaranteed global crystallographic solution.

Build the rank, screen the fit, select peak strength

All scores are heuristics. Peak coverage is weighted detected-profile coverage, not an intensity fraction or phase probability.

1 Positional evidence minus chance agreement

Let $W = \Sigma i\ wi$ and M be the one-to-one matched pairs. Coverage $C = \Sigma(i,j)\in M\ wi / W$.

Position score $P = \Sigma(i,j)\in M\ wi \cdot Qij / W - \Sigma i\ wi \cdot ci / W$.

$ci = \min(1, \Sigma j\ 1[||oi|-|pj|| < \tau] \cdot (2/\pi) \cdot \operatorname{asin}(\min(\tau/\max(|oi|,0.1),1)))$.

RMS = √mean(dij²) over matched pairs, without intensity weighting. Chance correction accounts for prediction density.

2 Scale preference

$S = \lambda \cdot \text{mean}_k[((s_k - 1)/0.05)^2]$, over visible fitted reciprocal-scale parameters.

One uniform 5% adjustment costs λ; 10% costs 4λ. Fixed mode costs zero. This adjustable preference is not a measured strain limit or a parameter-count correction.

3 Check additional weak maxima and unsupported predictions

Search unmatched prediction sites within r = max(0.10, window/spot width). Exclude strong-peak neighbourhoods; use one-to-one weak assignments. In FFT mode this separate weak detector uses σ = 0.65 bins, a 1.8 floor and local annular median/MAD contrast.

For assessable sites, weak qualities include distance and bounded local contrast. B = max(0, weak quality per predicted node − mean of 11 rotated-control values).

F = unsupported predictions / assessable predictions. Edge, masked or unreliable sites are unchecked rather than penalized. Missing predictions are not proof of absent reflections.

4 Adjusted rank

$R = P - S + \eta \cdot B - \eta \cdot F$. λ controls scale preference; η controls weak/missing-peak evidence. Default λ = 0.10 and η = 0.10.

5 Acceptance screen

Require ≥4 original strong matched profiles, directional spread $\max|\sin(\theta i - \theta j)| > 0.5$, $C > 0.30$, $P > 0.20$ and final $R > 0.20$.

Weak maxima change rank but do not count toward four strong profiles. Passing is compatibility evidence, not a confirmed phase.

AUTO STRENGTH / reuse one FFT

Build the strongest-first component pool with minimum strength set to zero. Other thresholds, smoothing, separation and profile cap remain active.

Each distinct component strength defines a possible retained peak set.

Fit every admissible distinct set

Sweep the admitted strengths, including cutoffs stricter than the manual value. Recompute matching and weak-site evidence for each set.

Do not stop at the first passing fit. If ≥4 profiles exist, do not explore subsets with fewer than four.

Select the best passing rank

Prefer sets with a passing candidate; among them maximize R. If none passes, keep the highest-ranked rejected result.

Equal ranks retain more profiles. Report selected strength, number of sets and trial scores.

Apply per ROI or per grid cell

The chosen set supplies the displayed profiles, classification and saved result. Selecting a threshold cannot manufacture an independent confirmation.

Cached reweighting

Changing λ or η can rerank stored candidates without repeating the FFT. It keeps the chosen profile set; rerun Auto analysis/grid to re-optimize the strength.

Interpret competing fits and periodic geometry

Passing candidates can describe the same evidence or separate evidence. Those cases produce different labels.

Any passing candidate?

No

No phase assignment

Fewer than 4 detected profiles → Insufficient evidence. Otherwise → Phase unresolved.

Choose the highest-ranked passing fit

Its matched profile IDs form the primary evidence set. Search other passing candidates for independent evidence beyond this set.

Independent second pattern?

Yes

Separate-evidence result

Require ≥3 new strong profiles, nonparallel spread >0.5 and >15% of total profile weight. Different phase → Two candidates; same phase → Multiple orientations.

No

Close competing phase?

Yes

Shared-evidence result

Rank gap <0.08 and >60% of primary IDs shared with another phase → Ambiguous. No separate population is established.

No

Otherwise: compatibility or unresolved

Primary weighted coverage >45% → compatible with that structure. Otherwise → Phase unresolved. The independent-evidence test takes precedence over near-tie ambiguity.

Additional shared-evidence ambiguity check

If a compatible leader and another passing phase match exactly the same profiles, but the leader uses more scale parameters or more unsupported prediction positions, retain Ambiguous.

PARALLEL CHECK / geometry only

Seed two reciprocal basis vectors from the 16 strongest profiles. Round observations to integer coefficients and fit the basis by weighted least squares (up to 3 rounds).

Search is bounded by basis length, area, angular spread and coefficients −4…4. No crystal structure is assigned.

Primary reciprocal grid

Require ≥4 matched profiles, nonparallel spread ≥0.5 and at least one coupled basis coefficient.

Geometry score = position-weighted coverage − chance allowance − 0.03. Primary passes with score >0.25 and coverage ≥50%.

Secondary reciprocal grid

A second grid must pass score >0.20 and coverage ≥30%, with ≥3 new strong profiles in nonparallel directions.

Rejected or duplicate support is not exposed as a detected secondary geometry. Identical primary evidence does not establish another pattern.

Dominance and downstream use

Dominance = primary coverage / (primary coverage + unique secondary weight fraction). A valid secondary is substantial when dominance <80%.

Geometry supplies an independent explanation and can trigger adaptive refinement. It does not choose the HCP/BCC/FCC phase winner.

Candidate-labelled image area is not a measured phase fraction.

Adaptive regions, cached fits and saved state

Every accepted grid cell runs the same numerical pipeline. Refinement changes the map partition, not the detector physics.

1 Create a breadth-first region queue

Start with an n×n uniform grid (2–24 per side). Analyse each region using the submitted settings; maintain the currently accepted map leaves.

Adaptive and children large enough?

No

Keep current leaf

No split if adaptive is off, a child is <32 px, or its physical width would fall below the chosen stopping limit.

2 Split uncertain or multiple-pattern regions

Unresolved, insufficient, ambiguous, two-candidate or multiple-orientation results request subdivision. A substantial geometry-only second pattern also requests subdivision.

Queue four children when size limits allow.

3 Optional “Check passing regions”

Even a passing parent can test four children. Keep them if a child supports a new phase/zone, a separate pattern, or if the children mix passing and unassigned results.

A new phase/zone must beat the parent family on that same child by ≥0.08 rank, or the parent family must be absent/below screen.

4 Commit a subdivision or retain the parent

All four merely lose evidence → retain the parent. Same interpretation without a split trigger → retain the parent. Accepted subdivisions replace the parent and can refine recursively.

Passing-parent trial children do not count as map leaves until accepted. Cancelled/incomplete trials leave that parent intact.

Map totals use current leaf areas; subdivided parents and rejected probes are excluded.

Reweight an existing map

In-memory compressed candidate pools retain explored fits, including rejected fits. Changing scale penalty λ or peak weight η recomputes ranks, labels, symbols and totals without another FFT.

Cell boundaries and selected detected profiles stay fixed. Reweighting is not a fresh scale/rotation/strength search; rerun the grid when those choices need optimizing again.

Basic preferences and linked folders

Browser localStorage remembers basic analysis/display settings and per-image diffraction choices. Folder links are stored separately in .tem-paths.json.

Preferences are not a backup of the source data or a saved analysis.

Reusable named regions

tem_workbench.project.json stores image identity, ROI coordinates, settings, name, tags and notes. Writes use a revision check and atomic replacement.

Reopening recomputes the analysis. The project does not embed an FFT bitmap. Hover previews are a cache, not the scientific record. Original images stay in their linked data locations.

Exports and provenance

Numerical result/grid exports preserve settings, observations, fit evidence, classifications, source pixel checksum and engine hash. PNG/GIF exports separately capture the rendered view.

The browser draws the same axes and overlays for original FFT and processed detector signal. Diagnostic exports identify the Processed view; changing display never changes the numerical fit.

This atlas is versioned alongside the source. A hash check flags relevant code changes for diagram review; regeneration does not infer algorithm changes automatically.